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Faithfulness and Attribution Benchmarks for Scientific Retrieval-Augmented Language Models

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ABSTRACT

Evaluating whether scientific RAG models remain faithful to retrieved evidence is difficult due to the lack of fine-grained benchmark datasets. We introduce SciAttribution, an expert-annotated benchmark of 5,000 scientific question-answer pairs with token-level source attribution annotations. We define automated faithfulness metrics measuring entity support ratios, logical entailment scores, and hallucination severity indices. Our evaluations reveal that commercial conversational models hallucinate in 38.6% of complex multi-paper synthesis queries despite retrieving correct sources. A remaining challenge is measuring conceptual extrapolations that are factually true in reality but absent from the provided context corpus.

Keywords— Natural Language Processing, Evaluation, Faithfulness, Benchmark Datasets

INTRODUCTION

Existing RAG evaluation benchmarks focus on trivia questions or news articles. Scientific reasoning requires verifying complex multi-sentence arguments with exact mathematical and chemical provenance.

METHODOLOGY

SciAttribution comprises 5,000 scholarly QA instances evaluated by 25 post-doctoral researchers. We define automated evaluation scripts utilizing fine-tuned NLI models to detect ungrounded entity insertions and false causal links.

RESULTS

Benchmarking 12 leading commercial and open-weight RAG pipelines revealed that standard systems hallucinate unsupported claims in 38.6% of scientific answers, underscoring the urgent need for constrained attribution decoders.

LIMITATIONS

Distinguishing between harmless background knowledge (e.g., 'DNA is a double helix') and harmful domain hallucinations remains subjective and challenging to automate without explicit context-boundary protocols.

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